

Idempotency: Make retries safe.

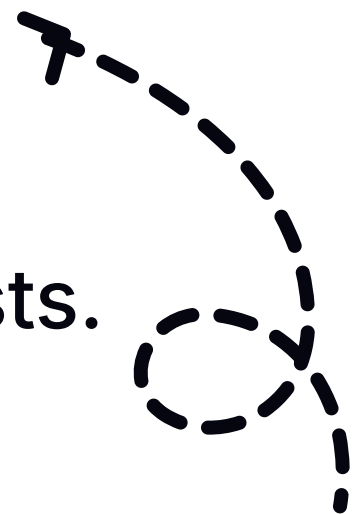
Networks fail. Clients retry.
Your system should be ready
for **both.**



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One customer. **Three** charges.

The client never sent three requests.
It sent one — and retried twice.



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SWIPE



03. WHAT ACTUALLY HAPPENED 3/8

The response died, not the write

`POST /payments → 201 • response lost`

Your server committed the charge. The connection dropped before the response came back. The client saw a timeout — and retried.

01 Attempt 1 — charge committed, response lost

02 Attempt 2 — retry, charged again

03 Attempt 3 — retry, charged again



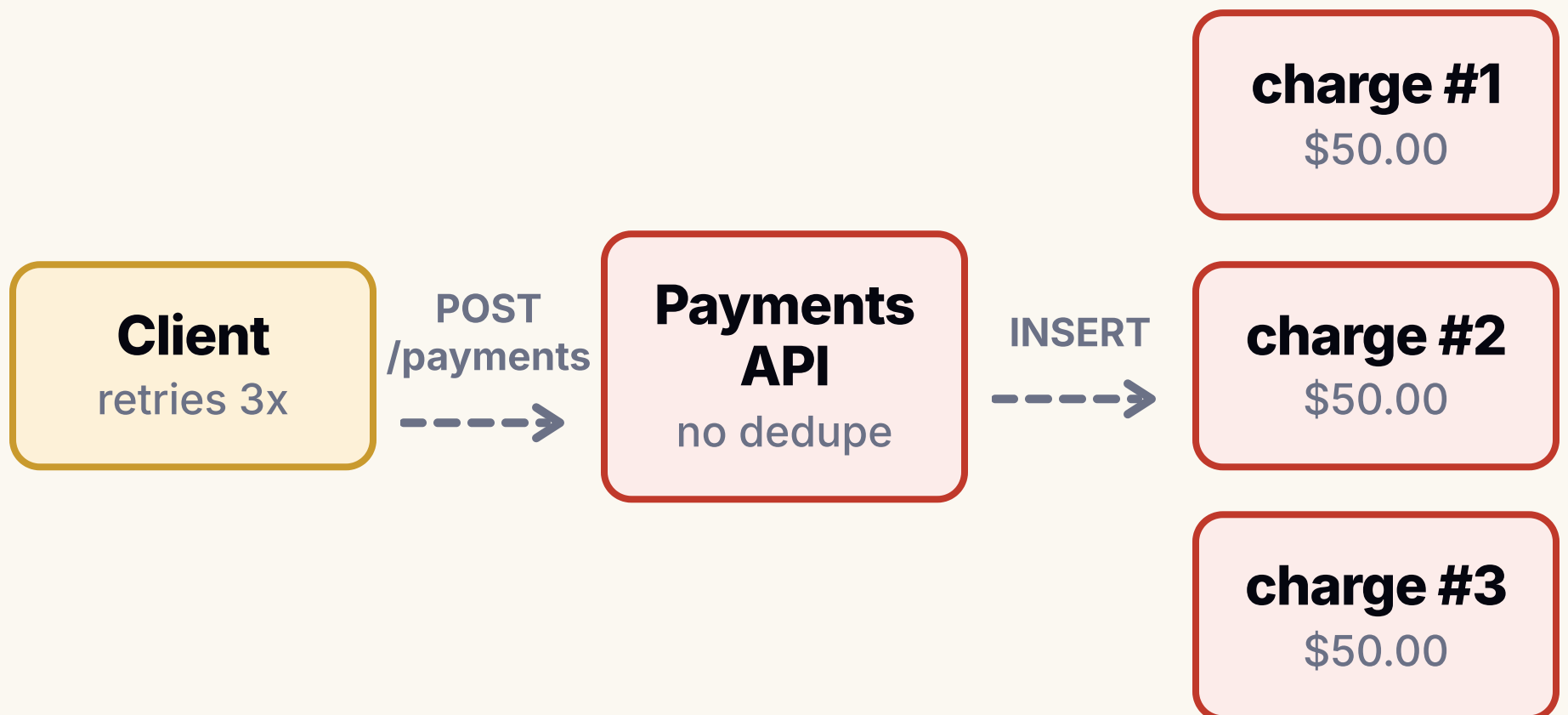
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The retry storm

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A timeout is not a failure.



A timeout tells you the response is missing. It never tells you the write didn't happen.



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05. THE FIX

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Give every attempt a name

One key per **intent** — not per attempt. The same key rides every retry.

WITHOUT A KEY

POST /payments



Every retry looks like a brand-new order.

WITH A KEY

Idempotency-Key: 7c9e-4f21



Every retry resolves to the same order.



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Claim the key first

payments.controller.js

JS

```
const key = req.header('Idempotency-Key');

// The unique index is the referee.
const claim = await db.query(
  `INSERT INTO idempotency_keys (key, status)
  VALUES ($1, 'in_progress')
  ON CONFLICT (key) DO NOTHING`, [key]);

if (!claim.rowCount) return replay(res, key);
```

→ If two retries race, exactly one wins the INSERT. The loser replays the stored response instead of charging again.



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Two retries, one row

schema.sql

SQL

```
CREATE TABLE idempotency_keys (  
  key          TEXT PRIMARY KEY,  
  status       TEXT NOT NULL,  
  response     JSONB,  
  created_at   TIMESTAMPTZ DEFAULT now()  
);
```

```
-- Two retries in the same millisecond both hit  
-- this index. Exactly one INSERT survives.
```

→ No lock. No queue. No distributed coordinator. One **PRIMARY KEY** does the arbitration for you.



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Takeaway

**A retry is not
a new request.
It is the **same intent**,
asking again.**

Design every write endpoint so the second attempt is boring.

Backend Engineering & System Architecture — daily.
Follow @peter-kekpe for the next one.



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